SimTIO: A Simulation-Grounded Multi-Agent LLM Framework for Compositional Traffic Intervention Optimization

Shuyang Li and Ruimin Ke (corresponding author)

Civil and Environmental Engineering
Rensselaer Polytechnic Institute
110 8th Street, Troy, New York 12180-3590, USA

Shuyang Li: lis36@rpi.edu; ORCID: 0009-0006-8588-2947.
Ruimin Ke: ker@rpi.edu.

Correspondence

Ruimin Ke
Civil and Environmental Engineering, Rensselaer Polytechnic Institute
110 8th Street, Troy, New York 12180-3590, USA
Email: ker@rpi.edu

CRediT authorship contribution statement

Shuyang Li: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing—original draft. Ruimin Ke: Supervision, Writing—review and editing.

Funding

This research was supported by the IBM–RPI Future of Computing Research Collaboration (FCRC).

Declaration of competing interest

The authors declare that they have no competing interests.

Author approval

All authors have reviewed and approved the manuscript and agree to its submission.